



南京航空航天大学
NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS

《大模型安全与知识增强》实验报告

题 目: 大模型知识编辑

院 系: 人工智能学院

专 业: 计算机科学与技术

姓 名: 熊攀

学 号: SX2524005

2026 年 5 月 24 日

一、实验要求

1.1 实验背景与目的

随着大语言模型（LLM）的广泛应用，模型知识过期或包含错误事实（幻觉）的问题日益凸显。重新训练或全量微调成本极高，而知识编辑（Knowledge Editing）技术允许我们在不显著改变模型其他行为的前提下，精准、快速地修改模型内部的特定知识。

实验目的：

1. 深入理解大语言模型中事实知识的存储机制。
2. 掌握并实践主流的知识编辑算法（如 ROME, MEMIT）。
3. 理解知识编辑的三大核心评估指标：编辑成功率（Efficacy）、泛化性（Generalization）与局部性/特异性（Locality）。

1.2 实验环境与工具准备

基础模型：Qwen2.5-0.5B 模型

核心框架：使用开源框架 EasyEdit

操作系统：Ubuntu 20.04.5 LTS

编程语言与版本：Python 3.10

显卡（GPU）：NVIDIA Tesla V100-PCIE-32GB

1.3 实验任务详解

Task 1: 基础环境搭建与基线测试 (Baseline Evaluation)

在不进行任何编辑操作的情况下，测试模型对特定过时知识或错误事实的回答。

任务内容：

构建包含 10 条事实更新的数据集（例如：“现任推特/X 公司的 CEO 是谁？”

-> 目标答案：“Linda Yaccarino”）。

编写推理脚本，记录原始模型的回答，证明模型在编辑前确实存在知识盲区或旧知识。

Task 2: 单条事实编辑实践 (Single Fact Editing using ROME)

ROME (Rank-One Model Editing) 是一种经典的通过修改前馈神经网络（FFN）特定层权重来实现单条知识编辑的方法。

任务内容：

基于 EasyEdit 框架，配置 ROME 算法参数。

将 Task 1 中的 10 条事实逐一进行编辑（每次重置模型权重）。

验证：输入提示词“推特现任 CEO 是”，观察模型是否输出“Linda Yaccarino”。

Task 3: 批量知识编辑实践 (Batch Editing using MEMIT)

MEMIT 算法在 ROME 的基础上进行了扩展，支持一次性向模型中注入成百上千条知识，同时保持较低的性能损耗。

任务内容：

选取开源知识编辑数据集 ZsRE 或 CounterFact 等其他数据集中的 500 条数据作为批量编辑集。

使用 MEMIT 算法对模型进行一次性批量知识注入。

记录批量编辑过程的显存占用和耗时情况。

Task 4: 综合评估 (Comprehensive Evaluation)

计算以下三个核心指标：

编辑成功率 (Efficacy, ES): 模型对直接编辑的 Prompt 是否输出了目标答案。

泛化性 (Generalization, PS): 模型对编辑事实的同义改写 Prompt 是否能输出目标答案。(例如：“谁目前在管理推特？”)

局部性 (Locality, NS): 模型对无关事实的回答是否受到了破坏。(例如：编辑了推特的 CEO，模型对“苹果的 CEO 是谁”的回答是否依然正确)。

Task 5: 进阶挑战 (Bonus - 选做)

跨语种泛化测试：用英文向模型注入事实（如 A 的首都是 B），测试模型用中文提问时（A 的首都是哪里？），是否能输出正确结果。

二、基线测试

首先是构建 10 条 Prompt，内容以 json 文件的格式存放在 prompts.json 中，随后将 prompt 输入到未编辑的模型中，运行 baselie.py 文件得到输出 original_model_answers.json，输出结果如下图所示：

```
Model: Qwen/Qwen2.5-0.5B
Device: cuda:0
Prompts: prompts.json
Output: original_model_answers.json

[1] Who is the current CEO of Twitter (X)?
Original answer: The current CEO of Twitter (X) is Jack Dorsey. He is the founder and CEO of Twitter, which was founded in 2006. Jack Dorsey has been the CEO of Twitter since 2011.

[2] Who is the current CEO of Starbucks?
Original answer: Starbucks is currently owned by Starbucks Corporation, which is a subsidiary of Starbucks Inc. The current CEO of Starbucks is Howard Schultz, who was appointed as the CEO in 2009.

[3] Who is the current Prime Minister of the United Kingdom?
Original answer: The current Prime Minister of the United Kingdom is Theresa May.

[4] Where is the headquarters of OpenAI located?
Original answer: The headquarters of OpenAI is located in San Francisco, California.

[5] Which team won the most recent FIFA World Cup?
Original answer: The 2018 FIFA World Cup was won by the South Korean team, South Korea.

[6] What is the official currency of Croatia?
Original answer: The official currency of Croatia is the HUF.

[7] What is the capital of Kazakhstan?
Original answer: Kazakhstan is a country in Central Asia. The capital of Kazakhstan is Astana.

[8] Which country is the newest member of NATO?
Original answer: The newest member of NATO is Albania.

[9] Which city will host the 2028 Summer Olympics?
Original answer: The 2028 Summer Olympics will be hosted in Tokyo, Japan.

[10] Who is the current Pope?
Original answer: Pope Francis is the current Pope.

Saved 10 original answers to original_model_answers.json
```

以问题"Who is the current CEO of Twitter (X)?"为例，模型输出的结果是"Jack Dorsey"，显然不符合事实，证明模型在编辑前确实存在知识盲区或旧知识。

三、RMOE 单条事实编辑

根据 baseline 得到的十条回答数据构建 10 条符合要求的测试集 testset.json，用来进行 RMOE 单条事实编辑，运行 edit_rome.py 文件，如下图所示：

```
Rewrite layer is 5
Tying optimization objective to 23
Recording initial value of v*
loss 4.466 = 4.466 + 0.0 + 0.0 avg prob of [ Pope Leo XIV] 0.012210184708237648
loss 3.273 = 3.093 + 0.18 + 0.0 avg prob of [ Pope Leo XIV] 0.04710788652300835
loss 2.617 = 2.415 + 0.202 + 0.0 avg prob of [ Pope Leo XIV] 0.09574103355407715
loss 2.235 = 1.64 + 0.594 + 0.001 avg prob of [ Pope Leo XIV] 0.19484125077724457
loss 1.467 = 1.157 + 0.309 + 0.001 avg prob of [ Pope Leo XIV] 0.31580880284309387
loss 1.153 = 0.903 + 0.249 + 0.001 avg prob of [ Pope Leo XIV] 0.407636433839798
loss 0.759 = 0.564 + 0.195 + 0.001 avg prob of [ Pope Leo XIV] 0.5720075368881226
loss 0.483 = 0.304 + 0.179 + 0.001 avg prob of [ Pope Leo XIV] 0.7405312061309814
loss 0.318 = 0.141 + 0.177 + 0.001 avg prob of [ Pope Leo XIV] 0.8693121075630188
loss 0.212 = 0.075 + 0.136 + 0.001 avg prob of [ Pope Leo XIV] 0.9283697009086609
loss 0.181 = 0.045 + 0.135 + 0.001 avg prob of [ Pope Leo XIV] 0.9562457203865051
loss 0.148 = 0.028 + 0.12 + 0.001 avg prob of [ Pope Leo XIV] 0.9725332856178284
loss 0.131 = 0.023 + 0.108 + 0.001 avg prob of [ Pope Leo XIV] 0.9776888489723206
loss 0.127 = 0.022 + 0.104 + 0.001 avg prob of [ Pope Leo XIV] 0.978603720664978
loss 0.117 = 0.02 + 0.096 + 0.001 avg prob of [ Pope Leo XIV] 0.9804916381835938
loss 0.107 = 0.017 + 0.09 + 0.001 avg prob of [ Pope Leo XIV] 0.9830330610275269
loss 0.102 = 0.015 + 0.087 + 0.001 avg prob of [ Pope Leo XIV] 0.9854148626327515
loss 0.095 = 0.012 + 0.082 + 0.001 avg prob of [ Pope Leo XIV] 0.9878716468811035
loss 0.088 = 0.01 + 0.078 + 0.001 avg prob of [ Pope Leo XIV] 0.9902475476264954
loss 0.082 = 0.008 + 0.073 + 0.001 avg prob of [ Pope Leo XIV] 0.9922086000442505
loss 0.075 = 0.007 + 0.068 + 0.001 avg prob of [ Pope Leo XIV] 0.9933779835700989
loss 0.063 = 0.006 + 0.056 + 0.001 avg prob of [ Pope Leo XIV] 0.9938083291053772
loss 0.043 = 0.006 + 0.036 + 0.001 avg prob of [ Pope Leo XIV] 0.9936487078666687
Delta norm: 27.263526916503906
Change in target norm: 6.815882205963135 to 28.092817306518555 => 21.276935577392578
Division Factor: 12.535191535949707
Right vector norm: 2.1749589443206787
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.5.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.5.mlp.down_proj.weight']
[10/10] Edited: The current Pope is -> Pope Leo XIV
Saved eval outputs to rome_edit_outputs.json
(EasyEdit) xp@ai04:~/homework/Knowledge_Editing$
```

得到输出结果 rome_edit_outputs.json，以问题"Who is the current CEO of Twitter (X)?"为例，进行编辑后模型输出的结果是"Linda Yaccarino"，说明知识编辑起到了作用。同时该文件会用于后续的综合评估。

```
{
  "subject": "Twitter (X)",
  "prompt": "The current CEO of Twitter (X) is",
  "direct_prompt": "The current CEO of Twitter (X) is",
  "target_true": "Jack Dorsey",
  "target_new": "Linda Yaccarino",
  "paraphrase_prompts": [
    "Who is the chief executive officer of X, formerly Twitter?"
  ],
  "neighborhood_prompts": [
    "The current CEO of Apple is"
  ],
  "neighborhood_ground_truth": [
    "Tim Cook"
  ],
  "post": {
    "direct": "Linda Yaccarino. The circumstances that led to Yaccarino's job as a financial analyst are as follow",
    "paraphrase": [
      ""
    ]
  }
}
```

四、MEMIT 批量知识编辑

选用 CounterFact 数据集中的前 500 条数据用来进行批量知识编辑，运行 edit_memit.py 文件，如下图所示：

```
LAYER 8
Writing 500 key/value pair(s) into layer 8
z error tensor(47.6860, device='cuda:0', grad_fn=<MeanBackward0>)
Retrieving covariance statistics for Qwen_Qwen2.5-0.5B @ model.layers.8.mlp.down_proj.
Computing Cov locally....
Loading cached data/stats/Qwen2.5-0.5B/wikipedia_stats/model.layers.8.mlp.down_proj.float32_mom2_100000.npz
0%|          | 0/1000 [00:00<?, ?it/s]
orig norm tensor(38.5167, device='cuda:0')
upd norm tensor(61.9479, device='cuda:0', dtype=torch.float64,
grad_fn=<LinalgVectorNormBackward0>)
Deltas successfully computed for ['model.layers.4.mlp.down_proj.weight', 'model.layers.5.mlp.down_proj.weight', 'model.layers.6.mlp.down_proj.weight', 'model.layers.7.mlp.down_proj.weight', 'model.layers.8.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.4.mlp.down_proj.weight', 'model.layers.5.mlp.down_proj.weight', 'model.layers.6.mlp.down_proj.weight', 'model.layers.7.mlp.down_proj.weight', 'model.layers.8.mlp.down_proj.weight']
Collecting post-edit outputs...
=== MEMIT Batch Editing Summary ===
Model: Qwen/Qwen2.5-0.5B
Edits: 500
Time (s): 1637.62
GPU memory before (MiB): {'allocated_mib': 1885.2734375, 'reserved_mib': 2000.0, 'max_allocated_mib': 1885.2734375, 'max_reserved_mib': 2000.0}
GPU memory after (MiB): {'allocated_mib': 1985.6484375, 'reserved_mib': 2760.0, 'max_allocated_mib': 4373.564453125, 'max_reserved_mib': 10576.0}
Saved stats to: /mnt/ssd1/xp/homework/Knowledge_Editing/memit_batch_stats.json
Saved eval outputs to: /mnt/ssd1/xp/homework/Knowledge_Editing/memit_edit_outputs.json
```

如图所示，批量编辑过程的显存占用和耗时情况记录在 memit_batch_stats.json 中，总用时在 27.2 分钟左右，显卡的最大显存占用约为 4.27GB，同时编辑后模型的所有输出全部保存在 memit_edit_outputs.json，用于后续的综合评估。

五、综合评估

根据 Task2 和 Task3 中得到的模型输出文件，编写综合指标评估脚本 evaluate.py，分别运行后得到的结果如下所示：

```
(EasyEdit) xp@ai04:~/homework/Knowledge_Editing$ python evaluate.py --input ./rome_edit_outputs.json --output ./rome_eval_results.json
=== Evaluation Summary ===
ES: 1.0000
PS: 0.8000
NS: 0.1000
Saved results to: /mnt/ssd1/xp/homework/Knowledge_Editing/rome_eval_results.json
```

```
(base) xp@ai04:~/homework/Knowledge_Editing$ python evaluate.py --input ./memit_edit_outputs.json --output ./memit_eval_results.json
=== Evaluation Summary ===
ES: 0.9720
PS: 0.7700
NS: 0.1800
Saved results to: /mnt/ssd1/xp/homework/Knowledge_Editing/memit_eval_results.json
```

并将相应的结果保存在 json 文件中，统计结果如下：

方法	样本数	ES	PS	NS
ROME	10	100.0%	80.0%	10.0%
MEMIT	500	97.2%	77.0%	18.0%

ES 和 PS 都很高说明知识编辑起到了很好的作用，并且相似问题回答也很准确，但是 NS 很低说明这同时会影响其他问题的回答，模型对无关事实的回答受到了破坏。

六、跨语种泛化测试

编写测试脚本 cross_languag_test.py，用英文问题“The capital of the UK is”和英文

回答” Paris”进行知识编辑，最后用中文提问“英国的首都是哪里”，如下图所示：

```
loss 0.096 = 0.01 + 0.085 + 0.001 avg prob of [ Paris] 0.9897688031196594
loss 0.079 = 0.004 + 0.074 + 0.001 avg prob of [ Paris] 0.9958190321922302
loss 0.064 = 0.002 + 0.061 + 0.001 avg prob of [ Paris] 0.9975687861442566
loss 0.052 = 0.002 + 0.049 + 0.001 avg prob of [ Paris] 0.998089611530304
loss 0.042 = 0.002 + 0.039 + 0.001 avg prob of [ Paris] 0.9981976747512817
Delta norm: 22.28775978088379
Change in target norm: 5.571939468383789 to 22.88199234008789 => 17.3100528717041
Division Factor: 10.530776977539062
Right vector norm: 2.1164400577545166
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.5.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.5.mlp.down_proj.weight']
=== After Edit ===
EN prompt: The capital of the UK is
EN answer: Paris and the capital of the UK is Paris. The capital of the United Kingdom is Paris and the capital of the United
ZH prompt: 英国的首都是哪里
ZH answer: ?
A. 伦敦
B. 巴黎
C. 里昂
D. 里昂
$
```

发现英文的回答能很好的输出是”Paris”，说明知识编辑成功，但是对中文问题的回答却没有成功，说明在这个模型上对跨语言的知识编辑的泛化能力并不好。